

What is CSS?

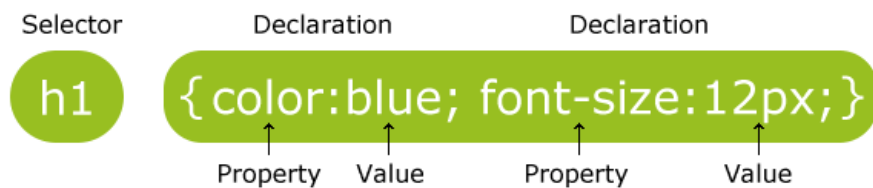
- Hypertext Markup Language or HTML is the basis for all Web pages.
- CSS stands for Cascading Style Sheets
- CSS describes how HTML elements are to be displayed on screen, paper, or in other media
- CSS saves a lot of work. It can control the layout of multiple web pages all at once
- External stylesheets are stored in CSS files
- Some of its features are as follows:
 - a. **In-built responsive designing.**
 - b. **Standard CSS.**
 - c. **Free to use.**

Why Use CSS?

- CSS is used to define styles for your web pages, including the design, layout and variations in display for different devices and screen sizes.

Syntax

- A CSS rule-set consists of a selector and a declaration block:



- The selector points to the HTML element you want to style.
- The declaration block contains one or more declarations separated by semicolons.
- Each declaration includes a CSS property name and a value, separated by a colon.
- A CSS declaration always ends with a semicolon, and declaration blocks are surrounded by curly braces.
- In the following example all <p> elements will be center-aligned, with a red text color:

Example

```
p {
  color: red;
  text-align: center;
}
```

CSS - Selectors

- CSS selectors are used to "find" (or select) HTML elements based on their element name, id, class, attribute, and more.

The element Selector

- The element selector selects elements based on the element name.
- You can select all <p> elements on a page like this (in this case, all <p> elements will be center-aligned, with a red text color):

Example

```
p
{
  text-align: center;
  color: red;
}
```

The id Selector

- The id selector uses the id attribute of an HTML element to select a specific element.
- The id of an element should be unique within a page, so the id selector is used to select one unique element!
- To select an element with a specific id, write a hash (#) character, followed by the id of the element.
- The style rule below will be applied to the HTML element with id="para1":

Example

```
#para1
{
  text-align: center;
  color: red;
}
```

The class Selector

- The class selector selects elements with a specific class attribute.
- To select elements with a specific class, write a period (.) character, followed by the name of the class.
- In the example below, all HTML elements with class="center" will be red and center-aligned:

Example

```
.center
{
  text-align: center;
  color: red;
}
```

CSS - Backgrounds

- The CSS background properties are used to define the background effects for elements.
- CSS background properties:
 - background-color
 - background-image
 - background-repeat
 - background-attachment
 - background-position

Background Color

- The background-color property specifies the background color of an element.
- The background color of a page is set like this:

Example

Body

```
{  
  background-color: lightblue;  
}
```

With CSS, a color is most often specified by:

- a valid color name - like "red"
- a HEX value - like "#ff0000"
- an RGB value - like "rgb(255,0,0)"

Example

```
h1  
{  
  background-color: green;  
}  
  
div  
{  
  background-color: lightblue;  
}  
  
p  
{  
  background-color: yellow;  
}
```

Background Image

- The background-image property specifies an image to use as the background of an element.
- By default, the image is repeated so it covers the entire element.
- The background image for a page can be set like this:

Example

```
body  
{  
  background-image: url("paper.gif");  
}
```

Example

```
body  
{  
  background-image: url("bgdesert.jpg");  
}
```

Example

```
body  
{  
  background-image: url("gradient_bg.png");  
  background-repeat: repeat-x;  
}
```

Example

```
body  
{  
  background-image: url("img_tree.png");  
}
```

```
background-repeat: no-repeat;
background-position: right top;
}
```

Background Properties

Property	Description
<u>background</u>	Sets all the background properties in one declaration
<u>background-attachment</u>	Sets whether a background image is fixed or scrolls with the rest of the page
<u>background-color</u>	Sets the background color of an element
<u>background-image</u>	Sets the background image for an element
<u>background-position</u>	Sets the starting position of a background image
<u>background-repeat</u>	Sets how a background image will be repeated

CSS - Text

Text Color

- The color property is used to set the color of the text.
- With CSS, a color is most often specified by:
 - a HEX value - like "#ff0000"
 - an RGB value - like "rgb(255,0,0)"
 - a color name - like "red"
- Look at CSS Color Values for a complete list of possible color values.
- The default color for a page is defined in the body selector.

Example

```
body {color:blue;} h1 {color:#00ff00;} h2 {color:rgb(255,0,0);}
```

For W3C compliant CSS: If you define the color property, you must also define the background-color property.

Text Alignment

- The text-align property is used to set the horizontal alignment of a text.
- Text can be centered, or aligned to the left or right, or justified.
- When text-align is set to "justify", each line is stretched so that every line has equal width, and the left and right margins are straight (like in magazines and newspapers).

Example

```
h1 {text-align:center;} p.date {text-align:right;} p.main {text-align:justify;}
```

Text Decoration

- The text-decoration property is used to set or remove decorations from text.
- The text-decoration property is mostly used to remove underlines from links for design purposes:

Example

```
a {text-decoration:none;}
```

It can also be used to decorate text:

Example

```
h1 {text-decoration:overline;}
h2 {text-decoration:line-through;}
h3 {text-decoration:underline;}
h4 {text-decoration:blink;}
```

CSS - Font

- The font shorthand property sets all the font properties in one declaration.
- The properties that can be set, are (in order): "font-style font-variant font-weight font-size/line-height font-family"
- The font-size and font-family values are required. If one of the other values are missing, the default values will be inserted, if any.

Syntax

font: font-style font-variant font-weight font-size/line-height
font-family|caption|icon|menu|message-box|small-caption|status-bar|initial|inherit;

Property Values

Property/Value	Description
font-style	Specifies the font style. Default value is "normal".
font-variant	Specifies the font variant. Default value is "normal".
font-weight	Specifies the font weight. Default value is "normal".
font-size/line-height	Specifies the font size and the line-height. Default value is "normal".
font-family	Specifies the font family. Default value depends on the browser.

Examples

```
p.ex1
{
  font: 15px arial, sans-serif;
}
```

```
p.ex2
{
  font: italic bold 12px/30px Georgia, serif;
}
```

CSS – Link

- Links can be styled with any CSS property (e.g. color, font-family, background, etc.).

```
a
{
  color: red;
}
```

- In addition, links can be styled differently depending on what **state** they are in.

- The four links states are:
- `a:link` - a normal, unvisited link
- `a:visited` - a link the user has visited
- `a:hover` - a link when the user mouses over it
- `a:active` - a link the moment it is clicked

Example

```
/* unvisited link */
a:link {
    color: red;
}

/* visited link */
a:visited {
    color: green;
}

/* mouse over link */
a:hover {
    color: hotpink;
}

/* selected link */
a:active {
    color: blue;
}
```

CSS – Table

Table Borders

- To specify table borders in CSS, use the border property.
- The example below specifies a black border for `<table>`, `<th>`, and `<td>` elements:

Example

```
table, th, td
{
    border: 1px solid black;
}
```

Table Width and Height

- Width and height of a table are defined by the width and height properties.
- The example below sets the width of the table to 100%, and the height of the `<th>` elements to 50px:

Example

```
table
{
    width: 100%;
}
```

```
th
{
  height: 50px;
}
```

Table Padding

- To control the space between the border and the content in a table, use the padding property on <td> and <th> elements:

```
th, td
{
  padding: 15px;
  text-align: left;
}
```

Hoverable Table

- Use the :hover selector on <tr> to highlight table rows on mouse over:

Example

```
tr:hover {background-color: #f5f5f5}
```

Table Properties

Property	Description
border	Sets all the border properties in one declaration
border-collapse	Specifies whether or not table borders should be collapsed
border-spacing	Specifies the distance between the borders of adjacent cells
caption-side	Specifies the placement of a table caption
empty-cells	Specifies whether or not to display borders and background on empty cells in a table
table-layout	Sets the layout algorithm to be used for a table

CSS – Outline

- The CSS outline properties specify the style, color, and width of an outline.
- An outline is a line that is drawn around elements (outside the borders) to make the element "stand out".
- dotted - Defines a dotted outline
- dashed - Defines a dashed outline
- solid - Defines a solid outline
- double - Defines a double outline
- groove - Defines a 3D grooved outline. The effect depends on the outline-color value
- ridge - Defines a 3D ridged outline. The effect depends on the outline-color value
- inset - Defines a 3D inset outline. The effect depends on the outline-color value
- outset - Defines a 3D outset outline. The effect depends on the outline-color value
- none - Defines no outline
- hidden - Defines a hidden outline

Example

```
p
{
  border: 1px solid black;
  outline-color: red;
}

p.dotted {outline-style: dotted;}
p.dashed {outline-style: dashed;}
p.solid {outline-style: solid;}
p.double {outline-style: double;}
p.groove {outline-style: groove;}
p.ridge {outline-style: ridge;}
p.inset {outline-style: inset;}
p.outset {outline-style: outset;}
```

CSS – Margin

- The CSS margin properties are used to generate space around elements.
- The margin properties set the size of the white space outside the border.
- With CSS, you have full control over the margins. There are CSS properties for setting the margin for each side of an element (top, right, bottom, and left).
- CSS has properties for specifying the margin for each side of an element:
 - margin-top
 - margin-right
 - margin-bottom
 - margin-left

Example

```
p {
  margin-top: 100px;
  margin-bottom: 100px;
  margin-right: 150px;
  margin-left: 80px;
}
```

CSS – Padding

- The CSS padding properties are used to generate space around content.
- The padding clears an area around the content (inside the border) of an element.
- With CSS, you have full control over the padding. There are CSS properties for setting the padding for each side of an element (top, right, bottom, and left).
- CSS has properties for specifying the padding for each side of an element:
 - padding-top
 - padding-right
 - padding-bottom
 - padding-left

Example

```
p {
  padding-top: 50px;
  padding-right: 30px;
  padding-bottom: 50px;
  padding-left: 80px;
}
```

CSS – Align

- To just center the text inside an element, use text-align: center;

Example

```
.center
{
  text-align: center;
  border: 3px solid green;
}
```

CSS – Image Gallery

- CSS can be used to create an image gallery.
- The following image gallery is created with CSS:

Example

```
<html>
<head>
<style>
div.img {
  margin: 5px;
  border: 1px solid #ccc;
  float: left;
  width: 180px;
}

div.img:hover {
  border: 1px solid #777;
}

div.img img {
  width: 100%;
  height: auto;
}

div.desc {
  padding: 15px;
  text-align: center;
}
</style>
</head>
<body>

<div class="img">
```

```

<a target="_blank" href="fjords.jpg">
  
</a>
<div class="desc">Add a description of the image here</div>
</div>

<div class="img">
  <a target="_blank" href="forest.jpg">
    
  </a>
  <div class="desc">Add a description of the image here</div>
</div>

<div class="img">
  <a target="_blank" href="lights.jpg">
    
  </a>
  <div class="desc">Add a description of the image here</div>
</div>

<div class="img">
  <a target="_blank" href="mountains.jpg">
    
  </a>
  <div class="desc">Add a description of the image here</div>
</div>

</body>
</html>

```

CSS - Image Opacity

Transparent Image

- The opacity property can take a value from 0.0 - 1.0. The lower value, the more transparent:

```

img {
  opacity: 0.5;
  filter: alpha(opacity=50);

```

Transparent Hover Effect

The opacity property is often used together with the :hover selector to change the opacity on mouse-over:

```

img
{
  opacity: 0.5;
  filter: alpha(opacity=50);

```

```
}
```

img: hover

```
{  
  opacity: 1.0;  
  filter: alpha(opacity=100);  
}
```

Example

```
<html>  
<head>  
<style>  
img  
{  
opacity: 0.3; filter: alpha(opacity=30);  
}  
</style>  
</head>  
<body>  
    
</body>  
</html>
```

CSS – Navigation bar

- Having easy-to-use navigation is important for any web site.
- With CSS you can transform boring HTML menus into good-looking navigation bars.
- A navigation bar needs standard HTML as a base.
- In our examples we will build the navigation bar from a standard HTML list.
- A navigation bar is basically a list of links, so using the and elements makes perfect sense:

Example

```
<ul>  
  <li><a href="default.asp">Home</a></li>  
  <li><a href="news.asp">News</a></li>  
  <li><a href="contact.asp">Contact</a></li>  
  <li><a href="about.asp">About</a></li>  
</ul>
```

Vertical Navigation bar Example

```
<!DOCTYPE html>  
<html>  
<head>  
<style>  
ul {  
  list-style-type: none;  
  margin: 0;  
  padding: 0;  
  width: 200px;
```

```

    background-color: #f1f1f1;
    border: 1px solid #555;
}
li a {
    display: block;
    color: #000;
    padding: 8px 16px;
    text-decoration: none;
}
li {
    text-align: center;
    border-bottom: 1px solid #555;
}

li:last-child {
    border-bottom: none;
}
li a.active {
    background-color: #4CAF50;
    color: white;
}
li a:hover:not(.active) {
    background-color: #555;
    color: white;
}
</style>
</head>
<body>

```

<h2>Vertical Navigation Bar</h2>

<p>In this example, we center the navigation links and add a border to the navigation bar.</p>

```

<ul>
  <li><a class="active" href="#home">Home</a></li>
  <li><a href="#news">News</a></li>
  <li><a href="#contact">Contact</a></li>
  <li><a href="#about">About</a></li>
</ul>
</body>
</html>

```

Horizontal Navigation bar example

```

<!DOCTYPE html>
<html>

```

```

<head>
<style>
ul {
  list-style-type: none;
  margin: 0;
  padding: 0;
  overflow: hidden;
  background-color: #333;
}
li {
  float: left;
}
li a {
  display: block;
  color: white;
  text-align: center;
  padding: 14px 16px;
  text-decoration: none;
}
li a:hover {
  background-color: #111;
}
</style>
</head>
<body>
<ul>
  <li><a class="active" href="#home">Home</a></li>
  <li><a href="#news">News</a></li>
  <li><a href="#contact">Contact</a></li>
  <li><a href="#about">About</a></li>
</ul>
</body>
</html>

```